Order-Aware Residual Dual Control for Risk-Budgeted Multi-Level Market Making

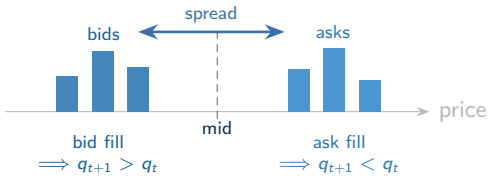
Maria Nesterova

Laboratory of Artificial Intelligence in Mathematical Finance (AIM Lab)
HSE University

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Market making: spread revenue and inventory exposure

A market maker earns the spread only when both sides of the book trade. One-sided fills leave inventory exposed to price movements.



State transition

$$q_{t+1} = q_t + F_t^b - F_t^a, \qquad W_t = b_t + q_t M_t$$

A quote can be profitable locally and still create a position that loses value after an adverse move.

Research problem. Maximize expected trading return while limiting expected discounted absolute inventory exposure.

Risk-budgeted market making as a CMDP

Controlled process

$$\mathcal{M}_C = (\mathcal{S}, \mathcal{A}, P, r, c, \rho_0, \gamma_R, \gamma_C, B), \qquad S_0 \sim \rho_0, \qquad S_{t+1} \sim P(\cdot \mid S_t, A_t).$$

Information and action

$$o_t = \big(\mathrm{LOB}_{t-63:t},\ \text{market signals},\ \text{private state}\big),$$
$$A_t = \pi_\theta(o_t) \in [-1, 1]^4, \qquad t = 0, \ldots, T-1.$$

Fill dynamics and instantaneous cost

$$q_{t+1} = q_t + F_t^b - F_t^a,$$
$$W_t = b_t + q_t M_t, \qquad c_t = \frac{|q_{t+1}|}{Q_{\mathrm{ref}}}.$$

Constrained control objective

$$J_R(\theta) := \mathbb{E}^{\pi_\theta}_{\rho_0, P}\left[\sum_{t=0}^{T-1} \gamma_R^t r_t\right], \qquad J_C(\theta) := (1-\gamma_C)\mathbb{E}^{\pi_\theta}_{\rho_0, P}\left[\sum_{t=0}^{T-1} \gamma_C^t c_t\right] \le B,$$

$$\theta^\star \in \arg\max_{\theta \in \Theta:\, J_C(\theta) \le B} J_R(\theta).$$

Information structure. The environment is Markov in S_t. TD3 receives o_t, a 64-snapshot history used as a Markov surrogate.

From CMDP to the residual dual feedback

$$\max_{\theta} \; J_R(\theta) \quad \text{subject to} \quad J_C(\theta) \le B$$

Lagrangian relaxation

$$\mathcal{L}(\theta, \lambda) = -J_R(\theta) + \lambda\big(J_C(\theta) - B\big), \qquad \lambda \ge 0.$$

$$\boxed{\min_{\theta} \sup_{\lambda \ge 0} \mathcal{L}(\theta, \lambda)}$$

$$\lambda_{k+1} = \Big[\lambda_k + \eta_\lambda\big(\widehat{J}_C(\theta_k) - B\big)\Big]_+ .$$

If the estimated expected cost exceeds the budget, dual ascent raises the price of risk.

Fixed backbone and proposed feedback

$$\kappa_0 = (1 - \gamma_C)\lambda_0, \qquad \lambda_{\mathrm{eff,m}} = \lambda_0 + \delta_m,$$

$$0 \le \delta_m \le \lambda_0.$$

The inherited learner keeps κ_0 fixed. The residual uses local pacing and resting orders, rather than only $\widehat{J}_C - B$.

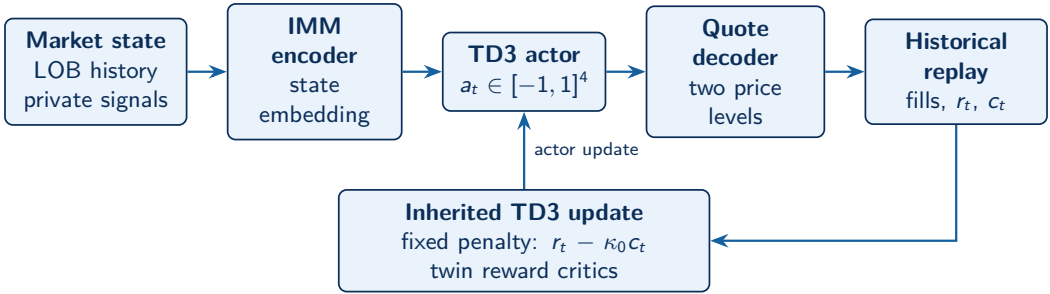
Neural saddle-point convergence is not assumed.

Research question. Can order-aware, time-local feedback improve control of the risk budget while retaining a stable fixed-penalty backbone?

Inherited fixed-penalty learning backbone

The order-aware controller starts from a seed-matched, already trained fixed-penalty checkpoint. The market encoder, TD3 learner and quote decoder are inherited.

INHERITED BACKBONE COMPONENTS



Observation. 64 recent top-five-book snapshots, market signals and private state.

Action. Center shift, spread and side allocation over two price levels.

Critics and inherited TD3 update

Value functions

$$Q_\kappa^\pi(s,a) = \mathbb{E}^\pi\left[\sum_{k\geq 0}\gamma_R^k\big(r_{t+k} - \kappa_0 c_{t+k}\big)\,\middle|\, s_t = s, a_t = a\right],$$

$$Q_C^\pi(s,a) = \mathbb{E}^\pi\left[\sum_{k\geq 0}\gamma_C^k c_{t+k}\,\middle|\, s_t = s, a_t = a\right], \qquad Q_C^+ = \mathrm{softplus}(Z_C) \geq 0.$$

TD3 targets

$$a' = \mathrm{clip}_{[-1,1]^4}\Big(\pi_{\theta^-}(\bar{s}_{t+1}) + \varepsilon_t\Big), \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2_{\mathrm{noise}} I).$$

$$y_\kappa = \tilde{r}_t + \gamma_R(1-d_t)\min_{i\in\{1,2\}} Q^-_{\kappa,i}(\bar{s}_{t+1}, a'),$$

$$y_C = c_t + \gamma_C(1-d_t)Q_C^{+,-}(\bar{s}_{t+1}, a').$$

Critic and actor losses

$$L_{\kappa,i} = \mathbb{E}_D\Big[(Q_{\kappa,i} - y_\kappa)^2\Big],$$

$$L_C = \mathbb{E}_D\Big[(Q_C^+ - y_C)^2\Big],$$

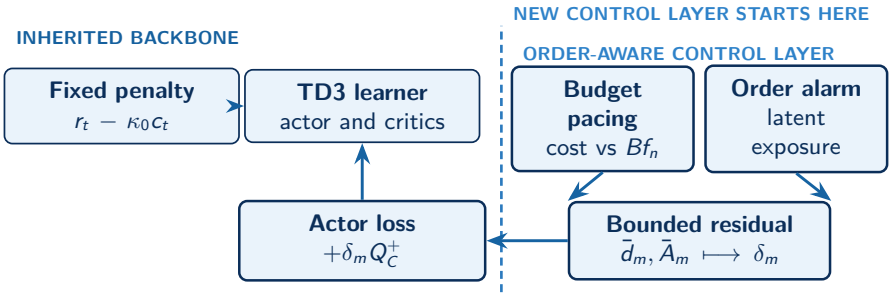
$$L_A^{(0)} = \mathbb{E}_D[-Q_{\kappa,1}(\bar{s}, \pi_\theta(\bar{s}))] + \beta_{BC} L_{BC}.$$

$$w^- \leftarrow (1-\tau)w^- + \tau w.$$

d_t is the terminal flag. The cost critic learns the unnormalized discounted sum; the factor $1-\gamma_C$ appears only in J_C.

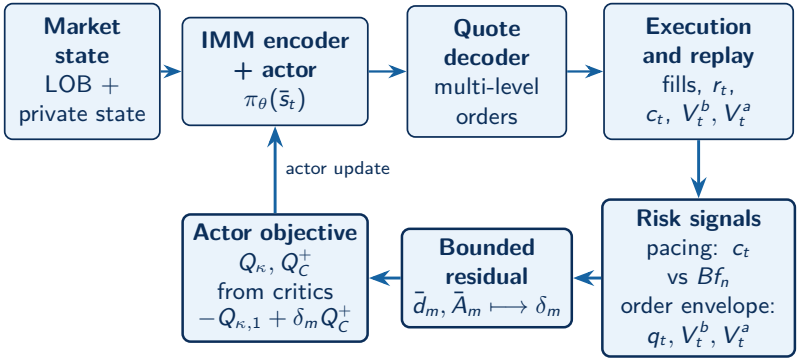
Where the contribution enters: order-aware control layer

The fixed-penalty backbone prices realized inventory with a fixed coefficient. The proposed layer adds a bounded residual driven by both realized and still resting exposure.



Novelty. The contribution begins when the fixed-penalty learner receives a stateful, order-aware residual risk signal.

Full architecture of the order-aware controller



Risk budget and its calibration

CMDP objective

$$\sup_{\theta \in \Theta} J_R(\theta; K) \qquad \text{subject to} \qquad J_C(\theta; K) \leq B.$$

Cost and normalized episode exposure

$$c_t = \frac{|q_{t+1}|}{Q_{\text{ref}}}, \qquad C_T = (1 - \gamma_C) \sum_{t=0}^{T-1} \gamma_C^t c_t.$$

$$\gamma_C = 0.999, \qquad Q_{\text{ref}} = 1.2455625 \text{ ETH}.$$

The constraint targets **expected discounted absolute inventory**, not a peak-position cap or a per-episode loss limit.

Training-only reference
fixed-penalty
mean exposure
$\overline{C}_{\text{train}} = 1.5178$

↓

20% reduction target
$B = 0.8 \cdot 1.5178 = \mathbf{1.2142}$

Calibration uses training data only; controller validation remains untouched.

Discount-consistent budget pacing

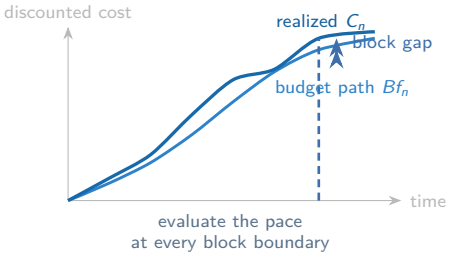
At each block, the controller compares incurred cost with a budget path under the CMDP discount factor.

Scheduled allocation

$$C_n = (1-\gamma_C)\sum_{t<n}\gamma_C^t c_t, \qquad B_n = Bf_n,$$

$$f_n = \frac{1-\gamma_C^{\min(n,T_0)}}{1-\gamma_C^{T_0}}, \qquad T_0 = 3600,$$

$$c_t = \bar{c} \implies \frac{C_n}{C_{T_0}} = f_n.$$



Smoothed blockwise residual

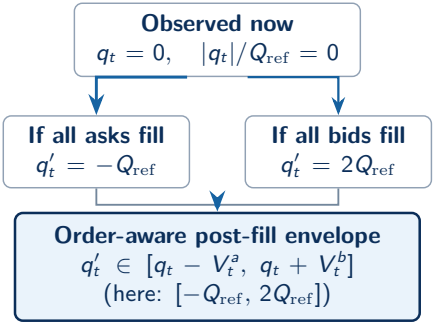
$$d_m = \frac{\Delta C_m - \Delta B_m}{\max(\Delta B_m, 10^{-8})}, \qquad \bar{d}_m = .75\bar{d}_{m-1} + .25\,\mathrm{clip}(d_m, -2, 2).$$

Updated every $H = 120$ decisions. The pacing signal follows the objective's discounting rather than an arbitrary linear schedule.

Pacing signal. Positive $\bar{d}_m$ means the latest block overspent its scheduled allocation.

Resting orders reveal latent exposure

Current inventory omits pending fills: resting orders can change the position before the next decision time.



$$\max_{q' \in [q_t - V_t^a,\ q_t + V_t^b]} |q'| = \max\{|q_t - V_t^a|,\ |q_t + V_t^b|\}.$$

$$\widetilde{c}_t = Q_{\mathrm{ref}}^{-1} \max\{|q_t + V_t^b|,\ |q_t - V_t^a|\},$$

$$g_t = \left[\widetilde{c}_t - \frac{|q_t|}{Q_{\mathrm{ref}}}\right]_+, \qquad A_t = \mathrm{clip}\left(\frac{g_t}{G} - 1, 0, 2\right).$$

$|q'|$ is convex, so its maximum on the feasible interval occurs at an endpoint. G is the 75th percentile of g_t on eligible training observations.

The bound considers fills of the current resting orders only, before new actions or other inventory changes. It measures **possible post-fill exposure**, not fill probability or a guarantee on future risk.

Residual control and actor pressure

The original fixed penalty stays fixed, and the residual is frozen within each actor step.

Bounded feedback law

$$
\begin{aligned}
z_m &= \bar{d}_m + \alpha \bar{A}_m, \\
b_m &= (1-\omega)\delta_m + \eta_P(\bar{d}_m - \bar{d}_{m-1}), \\
\widetilde{\delta}_{m+1} &= b_m + \eta_I z_m, \\
\delta_{m+1} &= \Pi_{[0,\lambda_0]}(\widetilde{\delta}_{m+1}).
\end{aligned}
$$

Conditional integration suppresses the integral term when it would push an already saturated residual outward. Thus

$$
\lambda_{\mathrm{eff,m}} = \lambda_0 + \delta_m, \qquad 0 \le \delta_m \le \lambda_0.
$$

New actor surrogate and its gradient

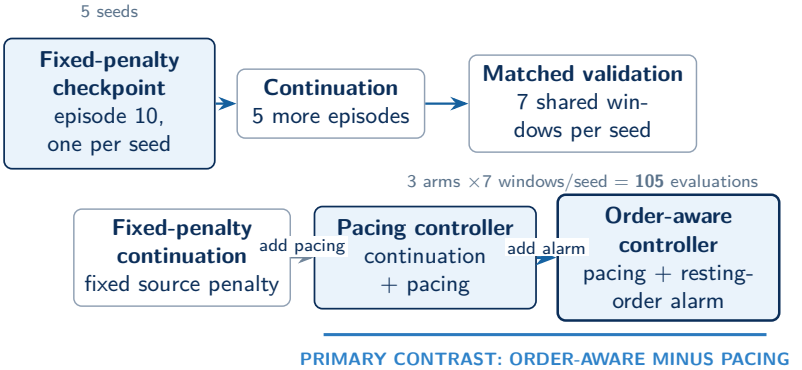
$$
\begin{aligned}
L_A(\theta) = \mathbb{E}_{\mathcal{D}}[&-Q_{\kappa,1}(\bar{s}, \pi_\theta(\bar{s})) \\
&+ \delta_m Q_C^+(\bar{s}, \pi_\theta(\bar{s}))\big] + \beta_{BC} L_{BC}.
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\theta L_A = \mathbb{E}_{\mathcal{D}}\Big[&- \nabla_\theta \pi_\theta(\bar{s})\, \nabla_a Q_{\kappa,1} \\
&+ \delta_m \nabla_\theta \pi_\theta(\bar{s})\, \nabla_a Q_C^+ \Big]_{a=\pi_\theta(\bar{s})} \\
&+ \beta_{BC} \nabla_\theta L_{BC}.
\end{aligned}
$$

Interpretation. Unlike dual ascent on $\widehat{J}_C - B$, this is a heuristic, block-local feedback law. Its residual scales the cost-gradient component that competes with the return-gradient component.

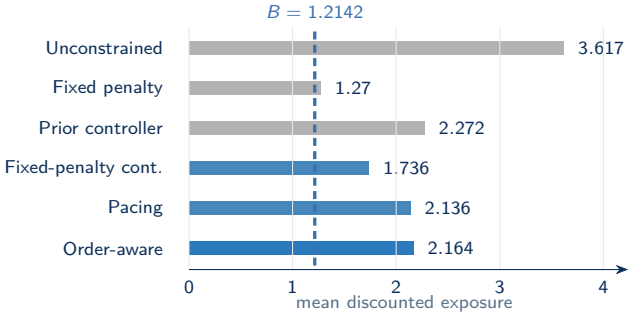
Matched continuation design

Every arm starts from the same seed-specific fixed-penalty checkpoint. The only intended difference between the pacing and order-aware controllers is the resting-order alarm.



Inference. Seed-window crossed bootstrap quantifies uncertainty in the matched mean-exposure contrasts.

Absolute budget target: no controller reaches feasibility



2.1642

Order-aware controller mean exposure

78.2%

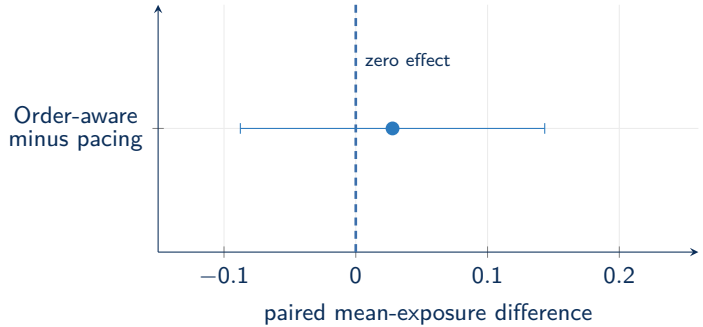
above the budget

The **fixed-penalty baseline** has the closest point estimate to the target.

Result. The absolute expected-exposure target is not achieved by any observed continuation.

Order-aware alarm: the primary contrast remains unresolved

The alarm effect is the order-aware minus pacing-controller mean-exposure contrast. Negative values would favor the order-aware controller.



Crossed 95% interval

$+0.0279$

$[-0.0876,\ 0.2178]$

The interval includes lower and higher expected exposure. It does not establish equivalence.

Result. The matched evaluation does not resolve the alarm effect.

Active feedback does not necessarily move the policy

The controller was active and stayed within its bounds. A policy change still requires a substantial additional gradient through the cost critic.

Observed feedback operation

Order alarm active	16.91%
Mean final residual	6.5810
Upper-cap occupancy	0.00%
Anti-windup events	221

Thus the negative result is not explained by a controller that never fired or was permanently clipped.

Path from residual to policy

$$\nabla_\theta \left(\delta_m Q_C^+\right) = \delta_m \left(D_\theta \pi_\theta\right)^\top \nabla_a Q_C^+ \quad \text{for fixed } \delta_m.$$

$$\frac{\left\| \delta_m (D_\theta \pi_\theta)^\top \nabla_a Q_C^+ \right\|}{\left\| (D_\theta \pi_\theta)^\top \nabla_a Q_{\kappa,1} \right\|} \approx 0.5\%\text{–}2.3\%.$$

Action saturation was 65.56%; the matched order-aware-minus-pacing interval still crosses zero.

Interpretation. Active feedback did not imply a large policy response. Weak cost gradients, discrete quote decoding and short continuation are hypotheses, not established causes.

Takeaways and next research step

What this work contributes

1. A bounded residual controller on top of a retained fixed penalty.

2. A calibrated live-order envelope that measures latent executable inventory.

3. A seed-window matched evaluation that separates controller operation from feasibility.

What the evidence says

1. The controller operates and preserves 74.06% of unconstrained activity.

2. No evaluated controller meets the absolute exposure budget.

3. The incremental alarm effect on mean exposure remains unresolved.

Next step: distributional exposure control

$$\sup_{\theta} J_R(\theta; K) \quad \text{subject to} \quad J_C(\theta; K) \leq B, \qquad \mathrm{CVaR}_{\zeta}(C_T) \leq B_{\mathrm{tail}}.$$

A distributional cost critic would make upper-tail inventory exposure observable, rather than relying only on its expected discounted value.

References: constraints and tail risk

Constraints and feedback

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